

Lecture 3:

Quick Intro to Classification and Log-linear Models

Reminder: Supervised Learning

- The task:
 - **Input:** training samples $x_1, x_2, x_3, \dots, x_n$ drawn from some distribution D , the learner is provided with their labels $y_1, \dots, y_n$
 - **Goal:** correctly predict the label of a new sample drawn from D
- Evaluation:
 - Take an annotated corpus and partition it as follows:



- Development data → for exploration; test data → for reporting results

Probabilistic Classification

- Two broad approaches to predicting classes y^*
 1. Joint / Generative: estimate a joint probabilistic model of the data
 - Usually assumes a functional form for $P(x/y)$, $P(y)$ and estimate parameters from the data
 - Use Bayes rule to compute $P(y/x)$
 - Inference:

$$y^* = \operatorname{argmax}_y P(y|x) = \operatorname{argmax}_y \frac{P(y)P(x|y)}{P(x)} = \operatorname{argmax}_y P(y)P(x|y)$$

- Advantages: learning is generally easy, smoothing is well-understood, a complete model of the data

Probabilistic Classification

2. Conditional / Discriminative (e.g., Logistic Regression)

- Only model conditional probability $P(y/x)$
- Prediction can be done directly:

$$y^* = \operatorname{argmax}_y P(y/x)$$

- Advantages: no need to model $P(x)$, easier to develop feature-rich models for $P(y/x)$

Maximum Likelihood Estimation

- The **likelihood** is the probability of the observed data given the parameters:

$$P(x, y; \theta)$$

x : the set of samples
 y : their corresponding labels
 θ : model parameters

- In discriminative models we often talk about the **conditional likelihood**:

$$P(y|x; \theta)$$

- The maximum estimator (MLE) is defined as $\theta_{MLE} = \operatorname{argmax}_{\theta} P(x, y; \theta)$
- And in discriminative models as $\theta_{MLE} = \operatorname{argmax}_{\theta} P(y|x; \theta)$

Maximum Likelihood Estimation

- Maximum likelihood estimation is a general estimation approach, commonly used in statistics
- In essence, it seeks to find the parameters under which the observed data will be the most likely

Simplest Generative Model: Naïve Bayes

- Represent each sample in a feature space $x_i \in \text{VALUES}^d$, where *VALUES* is a finite set of possible feature values
- Assume the label is discrete $y \in L$, where L is some finite set
- **Model:** the Naïve Bayes model is defined as

$$P(x, y) = P(y) \prod_{j=1}^d P(x^{(j)} | y)$$

- That is, features are assumed to be independent given the label

Simplest Generative Model: Naïve Bayes

- **Prediction:** for an example $x_i \in R^d$

$$y^* = \operatorname{argmax}_y P(y)P(x|y)$$

- As there are only so many values y can take, we just iterate over all of them and find the maximum

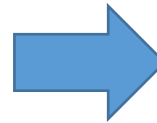
Example of a Naïve Bayes: Text Classification Task with Bag of Words Features

- Task: given text, predict what category it is from
- We assume that the topic is the label y , and represent the text as a count vector of the words in it
 - Each distinct wordform that may appear in the document is a feature
 - Values of features are usually indicators (0 or 1)

yossi likes apples

dina likes oranges

dana likes both apples and oranges



[1,0,0,1,1,0,0,0]

[0,1,0,1,0,1,0,0]

[0,0,1,1,1,1,1,1]

Features: *yossi, dina, dina, likes, apples, oranges, and, both*

- This works OK for text categorization if the topics are not too fine-grained

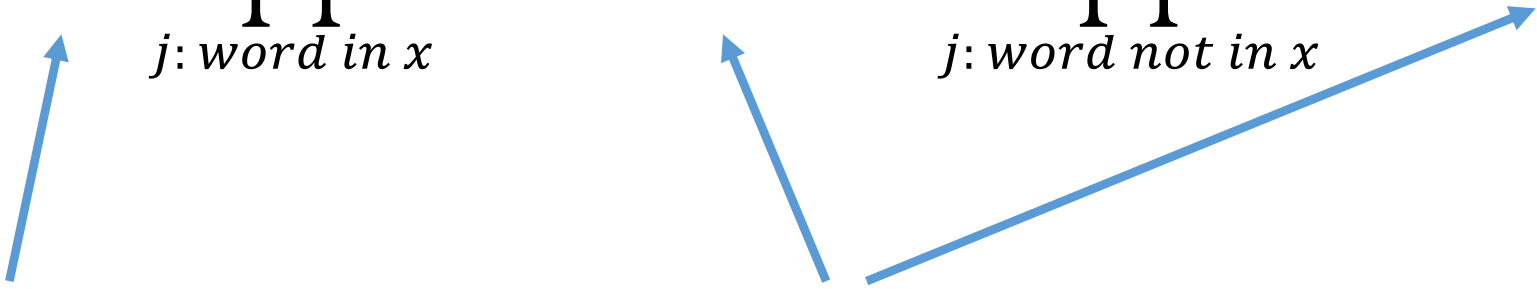
Prediction in Bag of Words Naïve Bayes

- Given a trained model, and a document x , we may predict the most likely y :

$$y^* = \operatorname{argmax}_y P(y)P(x|y) = \operatorname{argmax}_y P(y) \prod_{j=1}^d P(x^{(j)}|y) =$$

$$= \operatorname{argmax}_y P(y) \prod_{j: \text{word in } x} P(x^{(\text{word})} = 1|y) \prod_{j: \text{word not in } x} P(x^{(\text{word})} = 0|y)$$

prior



probability of a feature in a class

Learning in Bag of Words Naïve Bayes

- Parameters of the model:
 - Prior over the classes
 - Probability of a feature in a class
- For training, we assume N annotated examples and maximize their likelihood to find the maximum

$$L(\{x^{(i)}, y^{(i)}\}) = \prod_{i=1}^N P(x^{(i)}, y^{(i)}) = \prod_{i=1}^N P(y^{(i)}) \cdot P(x^{(i)}|y^{(i)}) = \prod_{i=1}^N P(y^{(i)}) \prod_{j=1}^d P(x_j^{(i)}|y^{(i)})$$

Learning in Bag of Words Naïve Bayes

- the Maximum Likelihood estimators turn out to be:

$$P(y = a) = \frac{|\{i : y^{(i)} = a\}|}{N}$$

$$P(x_j = 1|y = a) = \frac{|\{i : y^{(i)} = a \wedge x_j^{(i)} = 1\}|}{|\{i : y^{(i)} = a\}|}$$

- We need to apply some smoothing, obviously

Discriminative Approach

- Where there are complex features, generative approaches are more difficult to use
 - For example, highly correlated features
 - The identity of individual words (unigram features)
 - The identity of sequences of n words in the document (n-gram features)
- Many feature-based discriminative classification techniques out there, but log-linear models are still popular in the NLP community!

Text Classification

- Goal: classify documents into categories

... win the election ... *POLITICS*

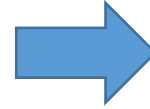
... win the game ... *SPORTS*

... see a movie ... *OTHER*

- Classically: based on bag of words in the document
- But other information sources are potentially relevant: document length, average word length, document's source, document layout

Feature Representation

Washington County jail *served*
11,166 meals last month - a figure
that translates to feeding some 120
people three times daily for 31 days

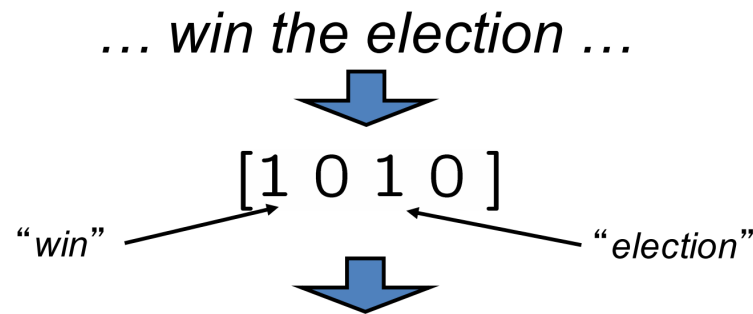


- Features are indicator functions which count the occurrences of certain patterns in the input
- Different feature values for every pair of input x and class y

context:jail = 1
context:county = 1
context:feeding = 1
context:game = 0
...
local-context:jail = 1
local-context:meals = 1
...
object-head:meals = 1
object-head:ball = 0

Block Notation

- We often think of the feature function as a mapping from a pair of input and label pair to a feature vector
 - In these cases, the feature vector will take a block form, as below



$$\begin{aligned}\phi(x, \textit{SPORTS}) &= [1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0] \\ \phi(x, \textit{POLITICS}) &= [0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0] \\ \phi(x, \textit{OTHER}) &= [0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0]\end{aligned}$$

Prediction

- In a linear model, each feature gets a weight
 - Feature function: ϕ
 - Weight vector: w

$$\begin{aligned}\phi(x, \textit{SPORTS}) &= [1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0] \\ \phi(x, \textit{POLITICS}) &= [0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0 \ 0] \\ \hline w &= [1 \ 1 \ -1 \ -2 \ 1 \ -1 \ 1 \ -2 \ -2 \ -1 \ -1 \ 1]\end{aligned}$$

- The prediction of y is the value that maximizes the score

$$\begin{aligned}\phi(x, \textit{SPORTS})^T w &= 1 * 1 + (-1) * 1 = 0 \\ \phi(x, \textit{POLITICS})^T w &= 1 * 1 + 1 * 1 = 2\end{aligned}$$



Prediction(x) = *POLITICS*

Log-linear Models

- Probabilistic model: use the scores as probabilities:

$$p(y|x; w) = \frac{\exp(w \cdot \phi(x, y))}{\sum_{y'} \exp(w \cdot \phi(x, y'))}$$

score(x,y,w)



- Prediction: straightforward. Compute $P(y/x)$ for each possible label, and select the maximum

Log-linear Models

- Learning: maximize the (log) conditional likelihood of training data

$$L(w) = \log \prod_{i=1}^n P(y_i | x_i; w) = \sum_{i=1}^n \log P(y_i | x_i; w)$$

$$w^* = \arg \max_w L(w)$$

Log-linear Models

- The conditional likelihood is concave, which means that we can maximize it using standard convex optimization techniques
 - Like gradient ascent or quasi-Newton methods
 - The gradient is given as:

$$L(w) = \sum_{i=1}^n \log P(y_i | x_i; w) \quad P(y | x; w) = \frac{e^{w \cdot \phi(x, y)}}{\sum_{y'} e^{w \cdot \phi(x, y')}}$$

$$\frac{\partial}{\partial w_j} L(w) = \sum_{i=1}^n \left(\phi_j(x_i, y_i) - \sum_{y'} P(y' | x_i; w) \phi_j(x_i, y') \right)$$

Total count of feature j
in correct candidates

Expected count of
feature j in predicted
candidates

Derivation

$$L(w) = \sum_{i=1}^n \log \left(\frac{e^{w \cdot \phi(x, y)}}{\sum_{y'} e^{w \cdot \phi(x, y')}} \right) =$$
$$\sum_{i=1}^n \left(w \cdot \phi(x, y) - \log \left(\sum_{y'} e^{w \cdot \phi(x, y')} \right) \right)$$

$$\frac{\partial L(w)}{\partial w_j} = \sum_{i=1}^n \left(\phi_j(x, y) - \frac{1}{\sum_{y'} e^{w \cdot \phi(x, y')}} \cdot \sum_{y''} \phi_j(x, y'') \cdot e^{w \cdot \phi(x, y'')} \right) =$$
$$\sum_{i=1}^n \left(\phi_j(x, y) - \sum_{y''} \phi_j(x, y'') \cdot \frac{e^{w \cdot \phi(x, y'')}}{\sum_{y'} e^{w \cdot \phi(x, y')}} \right) =$$
$$\sum_{i=1}^n \left(\phi_j(x, y) - \sum_{y''} P(y''|x) \cdot \phi_j(x, y'') \right)$$

Regularization

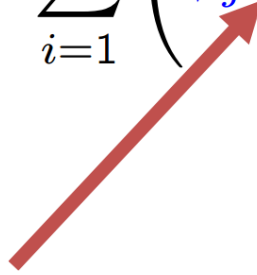
- The log-linear model doesn't have the same issue with zero counts as the generative model
- However, it may overfit the data by inflating w
 - If a certain feature appeared once with the class *SPORTS*, the model would have an incentive to place a very high weight for that feature and the label *SPORTS*
 - To combat this, we add a regularization term (often l_2 regularization)

$$L(w) = \sum_{i=1}^n \log p(y_i|x_i; w) - \frac{\lambda}{2} ||w||^2$$

Regularization: Modified Gradient

$$L(w) = \sum_{i=1}^n \left(w \cdot \phi(x_i, y_i) - \log \sum_y \exp(w \cdot \phi(x_i, y)) \right) - \frac{\lambda}{2} ||w||^2$$

$$\frac{\partial}{\partial w_j} L(w) = \sum_{i=1}^n \left(\phi_j(x_i, y_i) - \sum_y p(y|x_i; w) \phi_j(x_i, y) \right) - \lambda w_j$$



Total count of feature j
in correct candidates



Expected count of
feature j in predicted
candidates

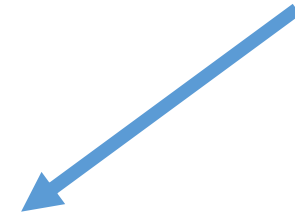


Big weights
are bad

SGD for Log-Linear Models and Perceptron

1. $w \leftarrow \vec{0}$
2. **for** $r = 1 \dots N_{iterations}$
3. **for** $i = 1 \dots N$
4. $w \leftarrow w + \eta(\phi(x_i, y_i) - \mathbb{E}_{\hat{y}}[\phi(x_i, \hat{y})])$
5. **return** w

In Perceptron, a
maximum rather
than expectation



Example: Named Entity Recognition

- The task: finding all the names mentioned in a text and classifying them into their types (e.g., *Location, Organization, Person, Other*)
- Example:

Citing high fuel prices, [ORG **United Airlines**] said [TIME **Friday**] it has increased fares by [MONEY **\$6**] per round trip on flights to some cities also served by lower-cost carriers. [ORG **American Airlines**], a unit of [ORG **AMR Corp.**], immediately matched the move, spokesman [PER **Tim Wagner**] said. [ORG **United**], a unit of [ORG **UAL Corp.**], said the increase took effect [TIME **Thursday**] and applies to most routes where it competes against discount carriers, such as [LOC **Chicago**] to [LOC **Dallas**] and [LOC **Denver**] to [LOC **San Francisco**].

Example: Regularization in NER

Because of regularization,
the more common prefixes
have larger weights even
though entire-word features
are more specific

Local Context

	Prev	Cur	Next
Word	at	Grace	Road
Tag	IN	NNP	NNP
Sig	x	Xx	Xx

Feature Weights

Feature Type	Feature	PER	LOC
Previous word	at	-0.73	0.94
Current word	Grace	0.03	0.00
Beginning bigram	Gr	0.45	-0.04
Current POS tag	NNP	0.47	0.45
Prev and cur tags	IN NNP	-0.10	0.14
Current signature	Xx	0.80	0.46
Prev-cur-next sig	x-Xx-Xx	-0.69	0.37
P. state - p-cur sig	O-x-Xx	-0.20	0.82
...			
Total:		-0.58	2.68

Concrete Case for Classification: Sentiment Analysis

Example #1: Movie Reviews

-  • Unbelievably disappointing
-  • Full of zany characters and richly applied satire, and some great plot twists
-  • This is the greatest screwball comedy ever filmed
-  • It was pathetic. The worst part about it was the boxing scenes.

Sentiment Analysis

Example #2: Product Reviews



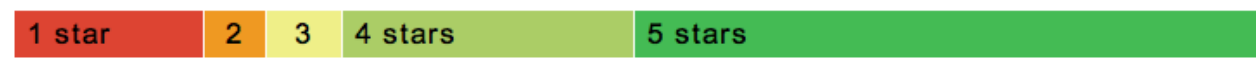
HP Officejet 6500A Plus e-All-in-One Color Ink-jet - Fax / copier / printer / scanner

\$89 online, \$100 nearby ★★★★★ 377 reviews

September 2010 - Printer - HP - Inkjet - Office - Copier - Color - Scanner - Fax - 250 sh

Reviews

Summary - Based on 377 reviews

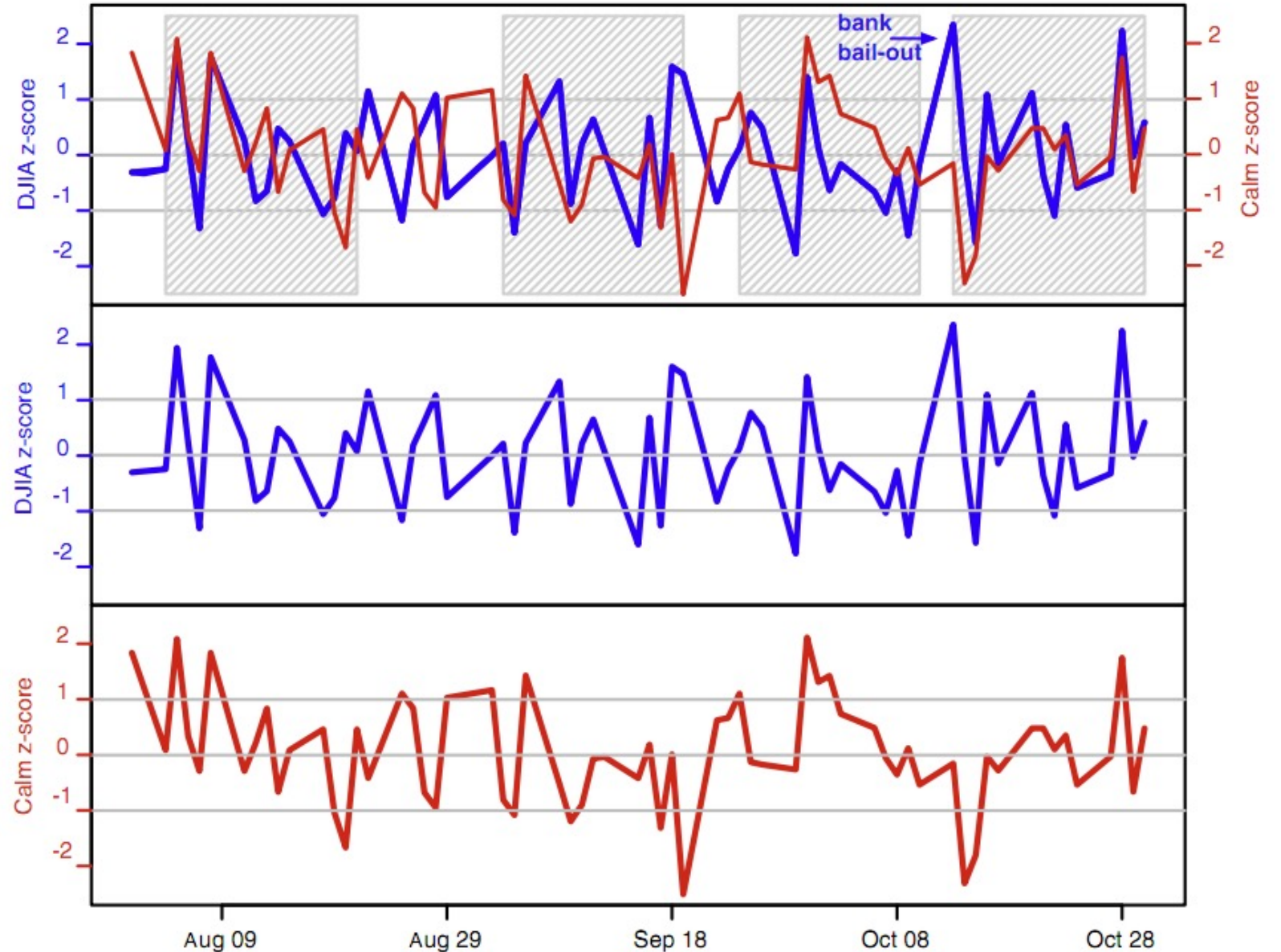


What people are saying

ease of use		"This was very easy to setup to four computers."
value		"Appreciate good quality at a fair price."
setup		"Overall pretty easy setup."
customer service		"I DO like honest tech support people."
size		"Pretty Paper weight."
mode		"Photos were fair on the high quality mode."
colors		"Full color prints came out with great quality."

- A Sentiment Analysis system called CALM applied to Twitter predicts the Dow Jones Industrial Average (DJIA) 3 days later
- Such algorithms are already used by hedge funds

Johan Bollen, Huina Mao, Xiaojun Zeng. 2011. [Twitter mood predicts the stock market](#), Journal of Computational Science 2:1, 1-8. 10.1016/j.jocs.2010.12.007.



Scherer Typology of Affective States

- **Emotion:** brief organically synchronized ... evaluation of a major event
 - *angry, sad, joyful, fearful, ashamed, proud, elated*
- **Mood:** diffuse non-caused low-intensity long-duration change in subjective feeling
 - *cheerful, gloomy, irritable, listless, depressed, buoyant*
- **Interpersonal stances:** affective stance toward another person in a specific interaction
 - *friendly, flirtatious, distant, cold, warm, supportive, contemptuous*
- **Attitudes:** enduring, affectively colored beliefs, dispositions towards objects or persons
 - *liking, loving, hating, valuing, desiring*
- **Personality traits:** stable personality dispositions and typical behavior tendencies
 - *nervous, anxious, reckless, morose, hostile, jealous*

Sentiment Analysis: Definition

- Sentiment analysis is the detection of **attitudes**
“enduring, affectively colored beliefs, dispositions towards objects or persons”
 1. **Holder (source)** of attitude
 2. **Target (aspect)** of attitude
 3. **Type** of attitude
 - From a set of types: *like, love, hate, value, desire*, etc.
 - Or (more commonly) simple weighted **polarity**: *positive, negative, neutral*, together with *strength*
 4. **Text** containing the attitude
 - Sentence or entire document

Sentiment Analysis

- Simplest task:
 - Is the attitude of this text positive or negative?
- More complex:
 - Rank the attitude of this text from 1 to 5
- Advanced:
 - Detect the target, source, or complex attitude types
- Sometimes not only complete sentences, but also their sub-parts receive a sentiment value
 - [The pizza was overall bad], but the toppings were fun! → overall negative

Sentiment Classification in Movie Reviews

- Is an IMDB movie review positive or negative?



when _star wars_ came out some twenty years ago
, the image of traveling throughout the stars has
become a commonplace image . [...]

when han solo goes light speed , the stars change
to bright lines , going towards the viewer in lines
that converge at an invisible point .

cool .



“ snake eyes ” is the most aggravating
kind of movie : the kind that shows so
much potential then becomes
unbelievably disappointing .

it's not just because this is a brian
depalma film , and since he's a great
director and one who's films are always
greeted with at least some fanfare .

Bo Pang, Lillian Lee, and Shivakumar Vaithyanathan. 2002. Thumbs up? Sentiment Classification using Machine Learning Techniques. EMNLP-2002, 79—86.

Bo Pang and Lillian Lee. 2004. A Sentimental Education: Sentiment Analysis Using Subjectivity Summarization Based on Minimum Cuts. ACL, 271-278

A Simple Classifier

- Log-linear or Naïve Bayes classifier
- Features:
 - The identity of the words
 - Possibly mark-up (e.g., hashtags in Twitter, headers in HTMLs)
- Features are often binary
 - Indicating whether a word appeared or did not appear in the document (bag of words)
 - Often works better for text classification than word frequency
 - The effect of frequency is non-linear

Error Analysis:

What makes reviews hard to classify?

- **Sarcasm:**

- Perfume review in *Perfumes: the Guide*:
 - “If you are reading this because it is your darling fragrance, please wear it at home exclusively, and tape the windows shut.”
- On *Automobile Steering Wheel Attachable Work Surface*:
 - “You wouldn’t believe how much more interesting my commute is now that I have something to do other than just stare out the window! I’m using it right now to post this review and I never”



- **Thwarted Expectations and Ordering Effects:**

- “This film should be brilliant. It sounds like a great plot, the actors are first grade, and the supporting cast is good as well, and Stallone is attempting to deliver a good performance. However, it can’t hold up.”

Negation in Sentiment Analysis

- One practice is to add NOT_ to every word between negation and following punctuation:

didn't like this movie , but I



didn't NOT_like NOT_this NOT_movie but I

Negation in Sentiment Analysis

- This is a very crude solution:
 - Explicit negation is only one way to reverse polarity
 - “He **failed** to convey the importance of his message”
 - Logical structure can be more complex
 - “I don’t think anyone could have done a better job”
 - Negation scope:
 - “I didn’t like the exposition in this otherwise excellent film”
- More recent approaches take the context (surrounding words) into account in the features
 - More about this later in the course